

A Privacy Budgeting Framework for Online Experimentation

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Firms perform online experiments with multi-armed bandits to personalize what consumers are shown while balancing exploration and exploitation. However, third-parties can infer consumers’ underlying segments from observing which banners, ads, or recommendations consumers receive. To control this inference, we propose a privacy risk budget that firms can set *ex ante* to bound such third party belief updating using differential privacy. To spend this privacy risk budget, we propose two strategies: a constant privacy risk strategy and a dynamic privacy risk strategy that spend privacy risk differently across visitor. We study how privacy risk budgets affect experimentation performance in two applications—website design and a recommendation system—under these strategies. For both strategies, we analytically find privacy risk budgets that optimally balance exploration and exploitation. We then extend the idea of an experiment-level privacy risk budget to a firm-wide privacy risk budget. We apply this firm-wide privacy risk budget in an empirical setting with 78 experiments. We find that the dynamic strategy is particularly valuable in longer and more complex experiments, and that optimizing the allocation of a firm-wide privacy risk budget across experiments substantially improves learning performance.

Keywords: differential privacy; privacy risk budget; multi-armed bandits; online experimentation.

1. Introduction

Firms routinely experiment with website banners, links, and recommendations to improve engagement and profitability. However, online experimentation also raises privacy concerns, particularly because third-parties can track what consumers are shown and use those exposures to infer their interests and behaviors (Goldfarb and Tucker 2011, Summers et al. 2016, Kim et al. 2018, Shaddy et al. 2026). These concerns translate into economic costs for firms: consumers increasingly opt out of tracking (Tucker 2014, Johnson et al. 2020, Wang et al. 2024, Miller and Skiera 2024) and place a premium on privacy-respecting brands (Moffett et al. 2025).

$$\xi_t = \log \left(1 - K + \frac{K}{\left(\frac{K \log(t)}{t} \right)^{1/3}} \right). \quad (16)$$

Under this schedule, the firm can protect visitors' privacy without degrading learning performance relative to an ε -greedy policy without privacy protection. However, to ensure that privacy risk never exceeds the predetermined budget γ in Equation (14), we cap the per-visitor privacy risk as follows:

$$\xi_t = \min \left\{ \gamma, \log \left(1 - K + \left(\frac{K}{\left(\frac{K \log(t)}{t} \right)^{1/3}} \right) \right) \right\}. \quad (17)$$

This cap implies that once the budget binds, the exploration and exploitation probabilities are fixed, and the privacy risk ξ_t remains set at γ , ensuring that the experiment-wide privacy guarantee is preserved. The cap effectively binds the extent of exploitation the bandit can engage in, and ensures that even after many visitors the bandit still explores and randomizes assignment, thus allowing plausible deniability.

In Figure 5, we visualize the privacy risk (left panel), the probability of exploitation (middle panel), and the probability of exploration (right panel) under the dynamic privacy risk strategy. Early visitors face lower privacy risk as the bandit explore more, while later visitors face higher privacy risk, as the bandit exploits more frequently. When $\xi_t > \gamma$, the privacy risk and induced exploration and exploitation probabilities are constant.

The introduction of the budget cap in Equation (17) creates an additional regret component relative to the standard ε -greedy bound: once the cap takes effect, the strategy is forced to randomize more than the uncapped schedule would prescribe late in the experiment, which increases regret. This strategy yields the following expected regret bound (see

strategy regime. The flexible regime allows the firm to choose, for each experiment, whether to use the constant or dynamic privacy budgeting strategy. By contrast, the constant regime restricts all experiments to use the constant strategy, and the dynamic regime restricts all experiments to use the dynamic strategy. The dashed horizontal line at zero indicates the random (lower bound) benchmark, equivalently the $\Gamma = 0$ case. The solid horizontal line indicates the upper bound γ^* benchmark. We use text annotations to highlight the difference in median clicks between the managerial heuristic and the regret-based allocation benchmark.

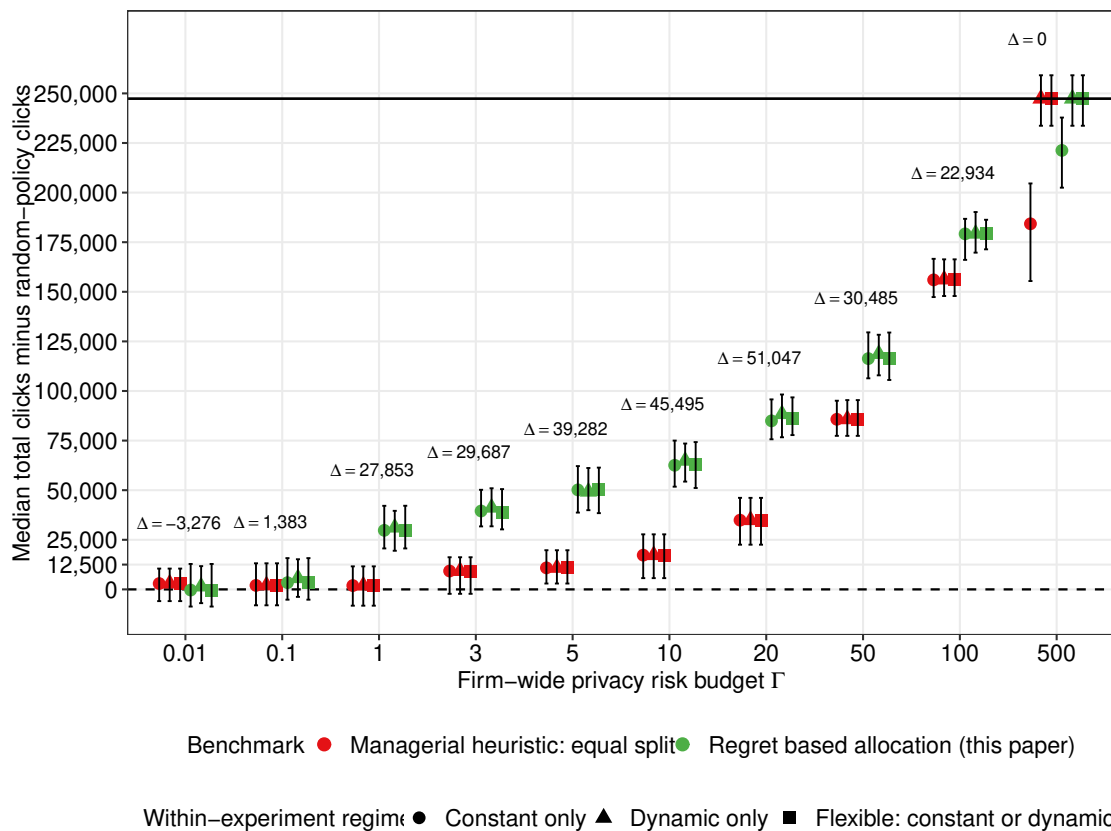


Figure 12 Median improvement in portfolio clicks relative to the random-policy baseline across firm-wide privacy budgets. Points report medians over simulation repetitions, and error bars show the interquartile range. Colors indicate the firm-wide allocation rule, while shapes indicate the within-experiment budgeting regime.

This privacy risk is controlled by the privacy-preserving assignment mechanism in Equation (13). Conditional on GA’s historical information, the measurement environment, and the GA channel being active, we bound the posterior-odds multiplier from observing the displayed arm:

$$R_A(a) \leq e^\gamma.$$

A.2. Privacy risk from behavior after experimental output

Second, if GA also observes behavior directly related to the experimental output, this behavior may create an additional source of posterior updating. For example, after observing that customer i was shown arm $A_i = a$, GA may also observe whether the customer clicked on that experimental arm. Conditional on the displayed arm, the experiment-specific behavioral-response factor is

$$R_{Y^E}(y^E) = \frac{\Pr(Y_i^E = y^E \mid H_i^{GA}, \Omega, G_i = 1, A_i = a, S_i = s)}{\Pr(Y_i^E = y^E \mid H_i^{GA}, \Omega, G_i = 1, A_i = a, S_i = s')}.$$

This factor measures how informative the customer’s experiment-specific response is once exposure has already occurred. For instance, if customers in segment s are more likely than customers in segment s' to click on arm a , then observing a click on the experimental output increases GA’s belief that the customer belongs to segment s .

A.3. The full posterior update

The model of privacy risk in experimentation is conditional on the GA observation channel being active. If $G_i = 0$, GA receives no current experimental signal, so the current experiment generates no output- or response-based posterior update. Conditional on $G_i = 1$, the posterior update depends on which experimental signals are observed under the measurement architecture Ω . If GA observes the displayed arm but not the experiment-specific response, then

$$\frac{\Pr(S_i = s \mid H_i^{GA}, \Omega, G_i = 1, A_i = a)}{\Pr(S_i = s' \mid H_i^{GA}, \Omega, G_i = 1, A_i = a)} = \frac{\Pr(S_i = s \mid H_i^{GA}, \Omega, G_i = 1)}{\Pr(S_i = s' \mid H_i^{GA}, \Omega, G_i = 1)} R_A(a).$$

If GA observes both the displayed arm and the raw experiment-specific response, then

$$\begin{aligned} & \underbrace{\frac{\Pr(S_i = s \mid H_i^{GA}, \Omega, G_i = 1, A_i = a, Y_i^E = y^E)}{\Pr(S_i = s' \mid H_i^{GA}, \Omega, G_i = 1, A_i = a, Y_i^E = y^E)}}_{\text{posterior odds}} \\ &= \underbrace{\frac{\Pr(S_i = s \mid H_i^{GA}, \Omega, G_i = 1)}{\Pr(S_i = s' \mid H_i^{GA}, \Omega, G_i = 1)}}_{\text{prior odds conditional on access}} \underbrace{R_A(a)}_{\text{experimental-output privacy risk}} \underbrace{R_{Y^E}(y^E)}_{\text{experiment-specific behavior privacy risk}}. \end{aligned}$$